

- 4 (a) State the conditions for the formation of a stationary wave.

.....

 [2]

- (b) Two radio transmitters emit vertically polarised electromagnetic waves of frequency 9.0×10^7 Hz as shown in Fig. 4.1.

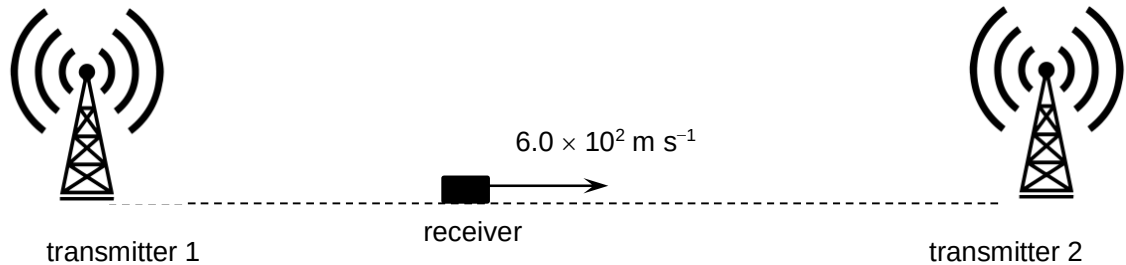


Fig. 4.1

- (i) Calculate the distance between 2 nodes in the stationary waves set up along the line joining the transmitters.

distance = m [2]

- (ii) A mobile receiver moves along the straight line joining the transmitters at a speed of $6.0 \times 10^2 \text{ m s}^{-1}$.

Calculate the rate at which the nodes are detected by the receiver.

rate = s^{-1} [2]

- (iii) Transmitter 2 is switched off and only transmitter 1 is still emitting electromagnetic radiation. At a location 10 km from transmitter 1, the receiver's reception fades temporarily to zero.

This is due to destructive interference between the direct beam of electromagnetic radiation and the beam reflected from a horizontal layer of ions formed in the atmosphere as shown in Fig. 4.2. The reflected beam undergoes a phase change of π radians upon reflection.

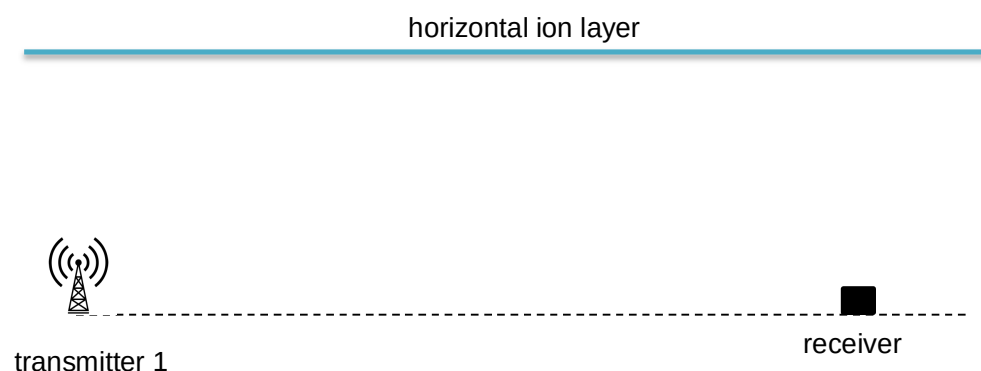


Fig. 4.2

Determine the minimum height of the reflecting ion layer.

minimum height = m [4]

5 (a) Using energy considerations distinguish between electromotive force (e.m.f.) and